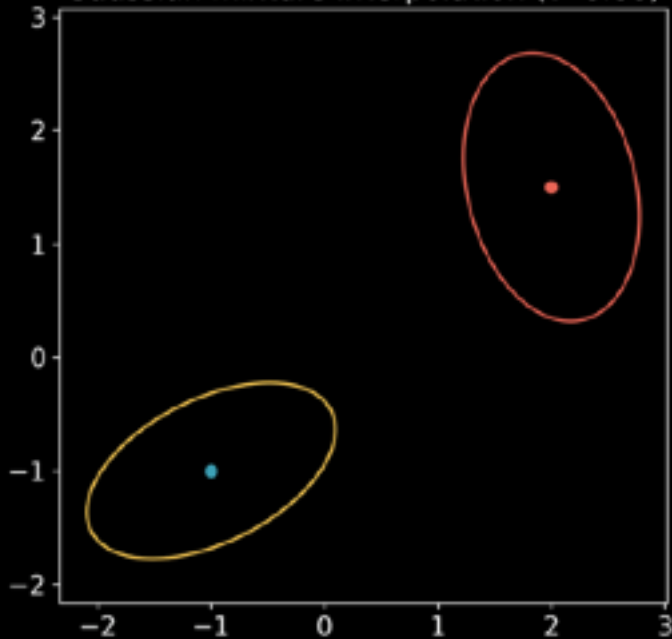
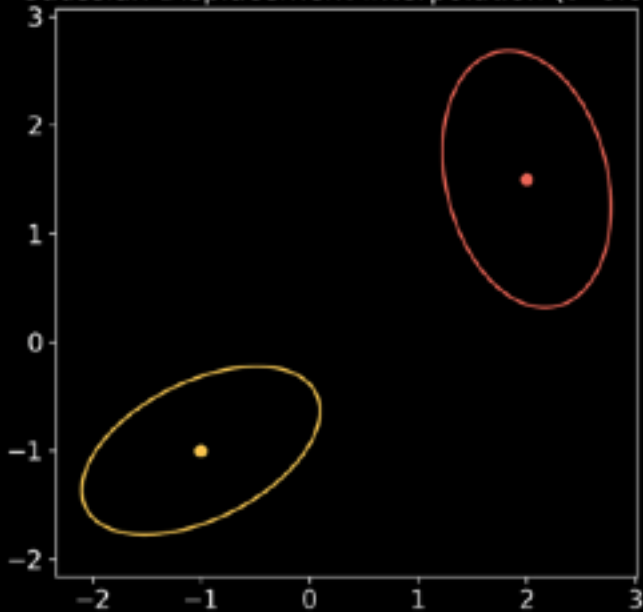


Gaussian mixture interpolation ($t=0.00$)



Gaussian Displacement interpolation ($t=0.00$)



Dynamic OT Formulation: Benamou-Brenier

$$W_2^2(\alpha, \beta) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \|v_t(x)\|^2 \rho_t(x) \, dx \, dt$$

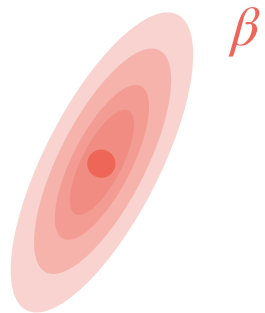
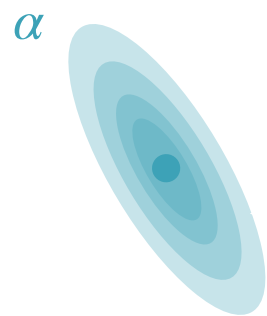
where ρ_t, v_t : measure, vector field. Subject to:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0 \quad (\text{continuity equation})$$

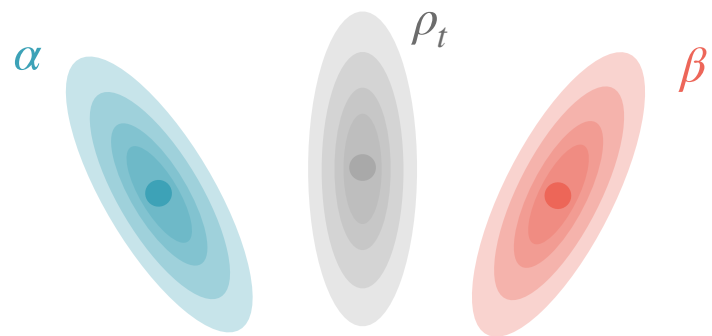
$$\rho_0 = \alpha, \rho_1 = \beta \quad (\text{boundary conditions})$$

locations

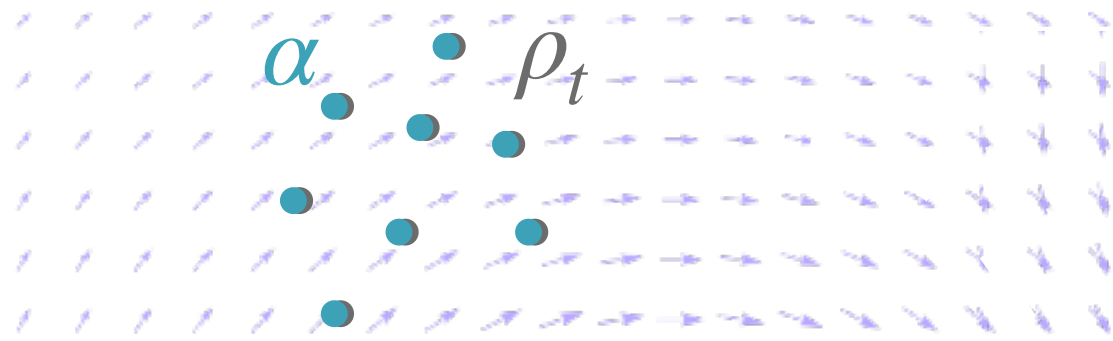
masses



min-cost (soft) matching



shortest-path (geodesic)



α

ρ_t

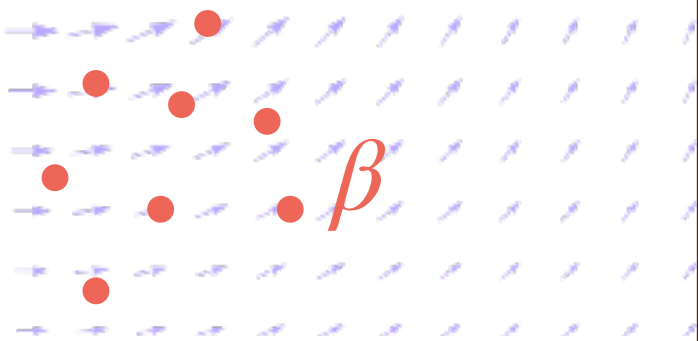
$\rho_t(x)$: measure at time t

$\{v_t, \rho_t\}_{t=0}^1$ defines a path in $\mathcal{P}(\mathcal{X})$

path length: $\int$ of velocity through time

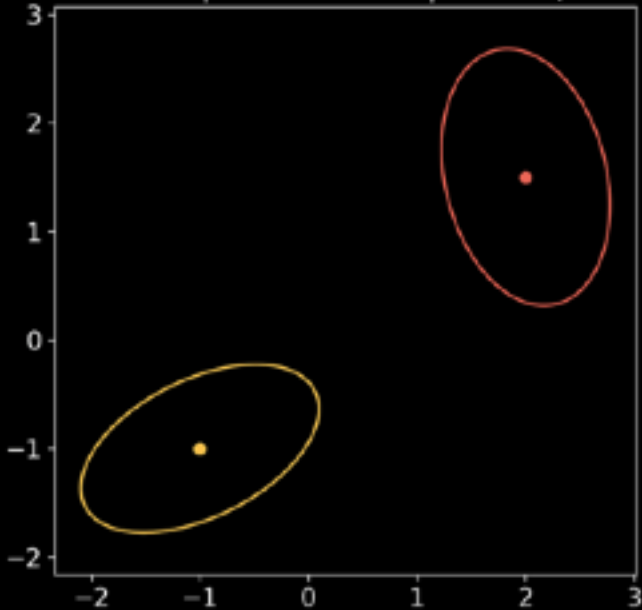
W_2 : length of shortest path

$v_t(x)$: time-varying vector field

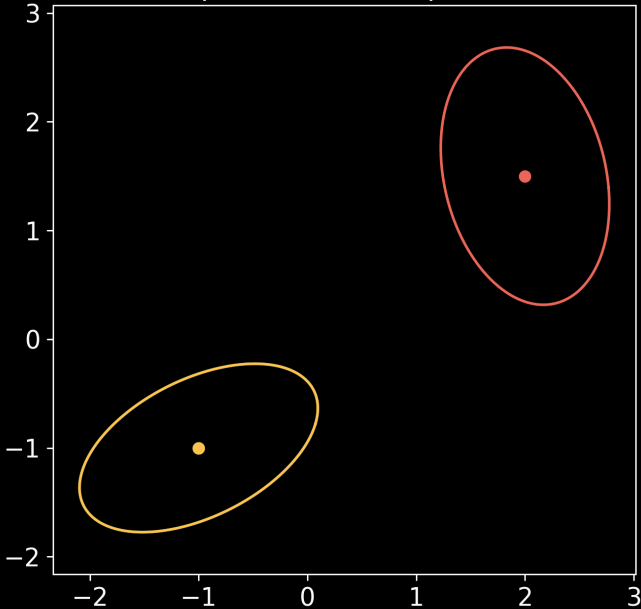


β

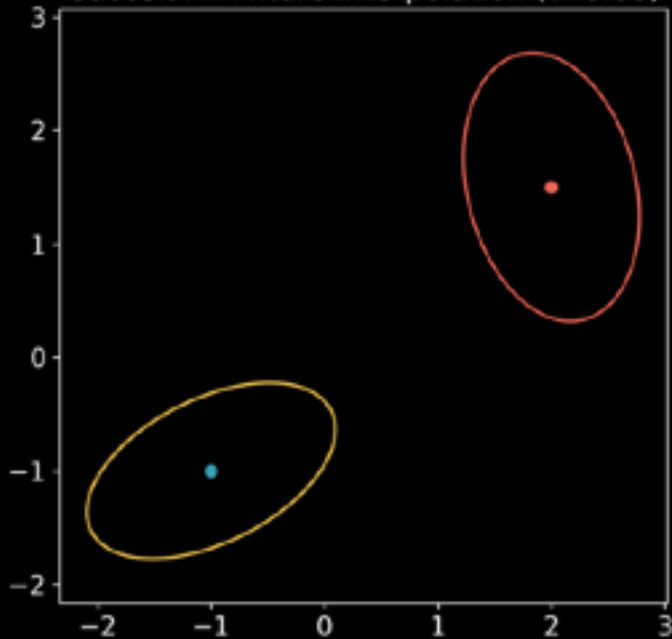
Gaussian Displacement interpolation ($t=0.00$)



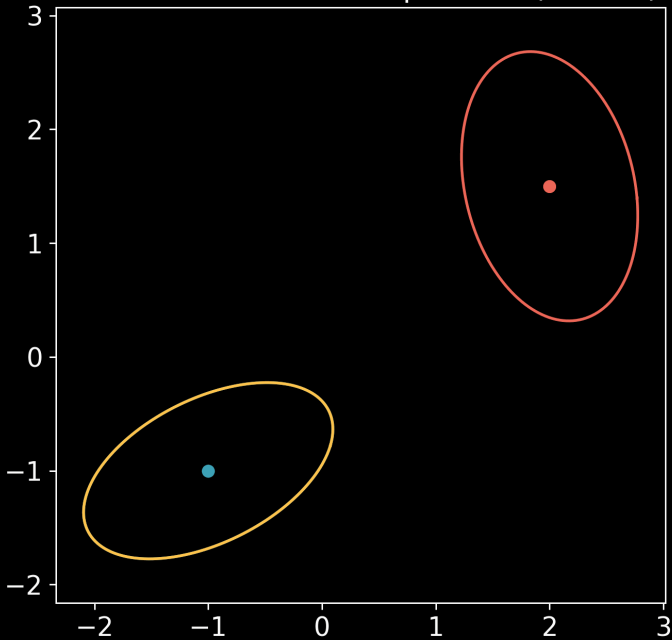
Gaussian Displacement interpolation ($t=0.00$)



Gaussian mixture interpolation ($t=0.00$)



Gaussian mixture interpolation (t=0.00)

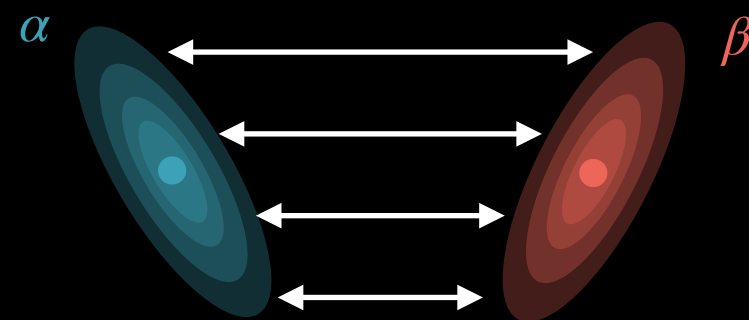


Last Time

Static vs Dynamic OT comparison

Static (Kantorovich)

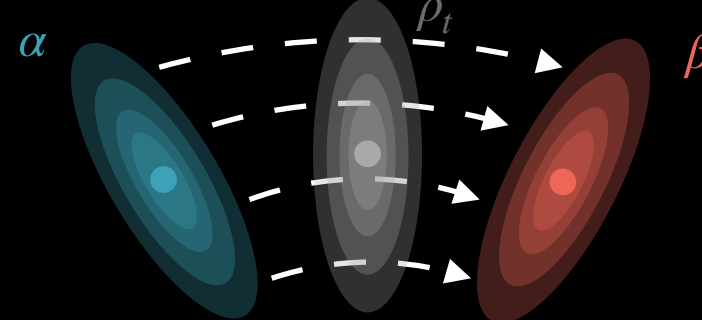
- Defined by couplings $\alpha \rightarrow \beta$
- Transport is one-shot, static



- Find coupling minimizing distance
- OT as **min-cost (soft) matching** between distributions

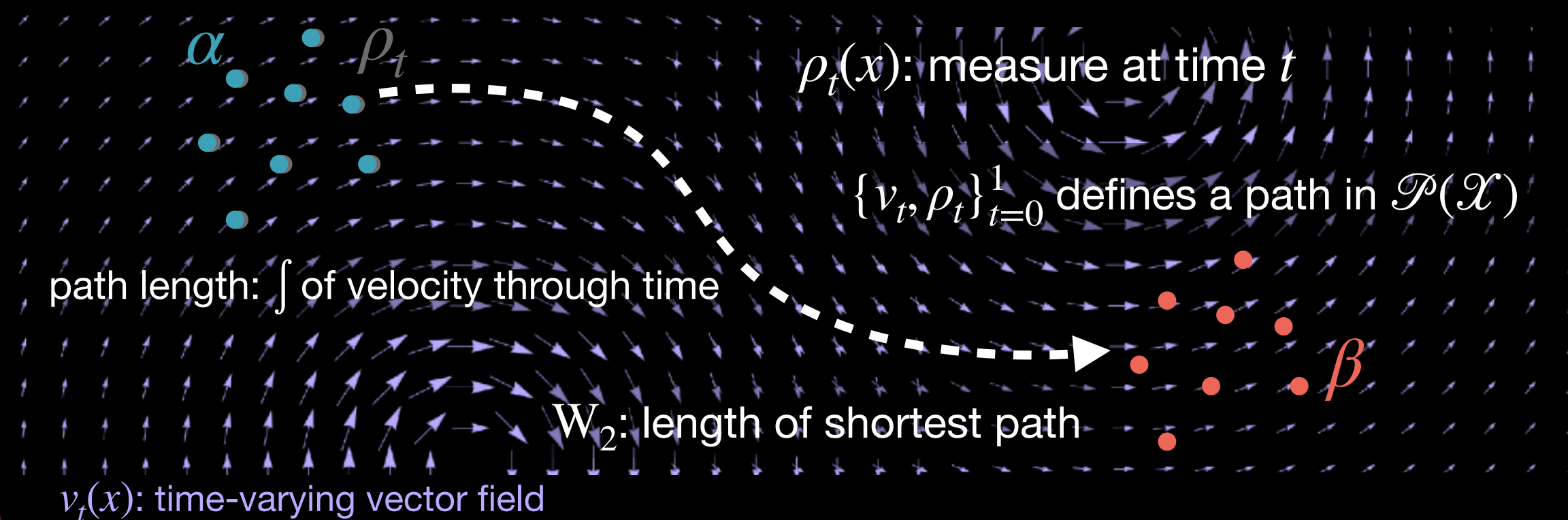
Dynamic

- Considers flows ρ_t, v_t transporting $\alpha \rightarrow \beta$
- Transport is time-dependent, dynamic



- Find flow minimizing energy
- OT as **shortest-path (geodesic)** in probability space

Benamou Brenier Intuition



Dynamic OT Formulation: Benamou-Brenier

$$W_2^2(\alpha, \beta) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \|v_t(x)\|^2 \rho_t(x) dx dt$$

where ρ_t, v_t : measure, vector field. Subject to:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0 \quad (\text{continuity equation})$$

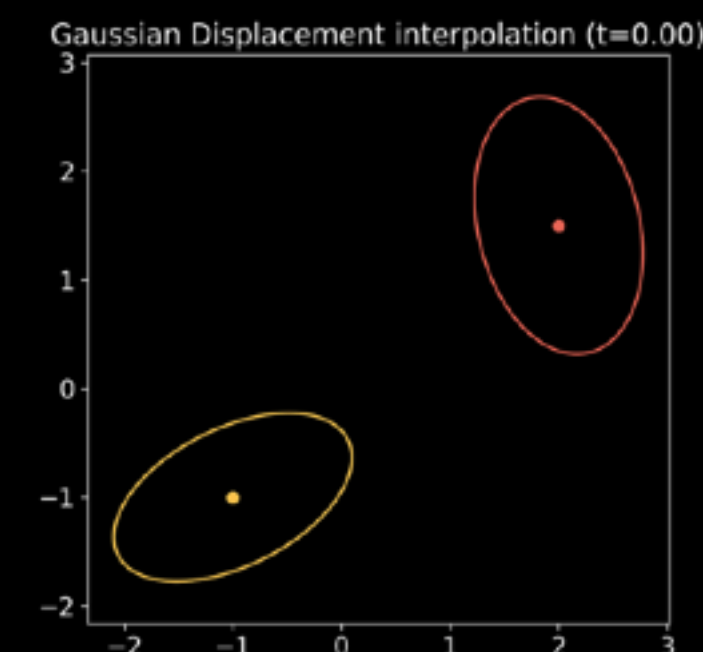
$$\rho_0 = \alpha, \rho_1 = \beta \quad (\text{boundary conditions})$$

Displacement vs Mixture Interpolation

Displacement Interpolation:

$$\rho_t^{\text{disp}} = [(1-t)\text{Id} + tT]_{\#} \alpha$$

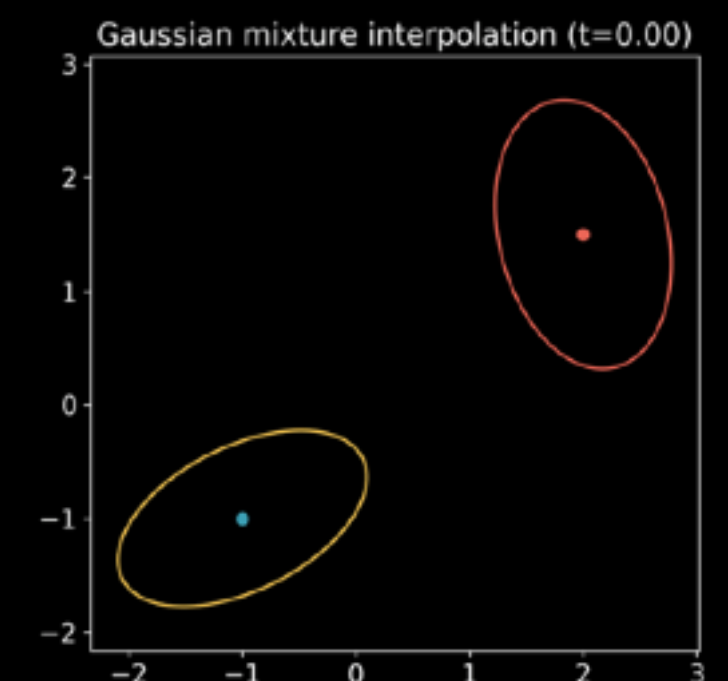
Interpolates **locations** of points



Mixture Interpolation:

$$\rho_t^{\text{mix}} = (1-t)\alpha + t\beta$$

Interpolates **masses** of points



Today